Classification or Regression?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One connected task-type and metric record.

Explain targets, trace category/error checks, repair code-as-quantity confusion and define a user-need test.

Keep one habit: Keep target meaning, units, pairing, boundary, per-case errors and bounded conclusions together.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

Features, Labels, Targets

My remembered idea, example and correction

MAIN LESSON · SECTION 12

W1 · Explain the mechanism

Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main duration pairs actual [8,10,14]/estimated [9,11,12], slow>10: classes [0,0,1]/[0,1,1], 2/3 match; errors [1,1,2] minutes, MAE 4/3≈1.33 minutes. W2 actual 14/estimate 12: both slow 1, error 2 minutes. Picture heights [4,7,9]/[5,6,10] cm: all errors 1 cm, MAE 1; categories 2/3 at >6, 3/3 at >7 with unchanged estimates.

  Category codes do not become quantities; zip assumes equal-length same-case order, units must match.

- At dispatch, identify the task for predicting route type coded 1 or 2, duration in whole minutes, and whether duration exceeds 10 minutes. Explain your choices.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main duration pairs actual [8,10,14]/estimated [9,11,12], slow>10: classes [0,0,1]/[0,1,1], 2/3 match; errors [1,1,2] minutes, MAE 4/3≈1.33 minutes. W2 actual 14/estimate 12: both slow 1, error 2 minutes. Picture heights [4,7,9]/[5,6,10] cm: all errors 1 cm, MAE 1; categories 2/3 at >6, 3/3 at >7 with unchanged estimates.

  Category codes do not become quantities; zip assumes equal-length same-case order, units must match.

- For actual 14 minutes and estimate **12 minutes**, calculate **absolute error** and both slow categories using greater than 10. Explain why a correct category does not mean an exact duration.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main duration pairs actual [8,10,14]/estimated [9,11,12], slow>10: classes [0,0,1]/[0,1,1], 2/3 match; errors [1,1,2] minutes, MAE 4/3≈1.33 minutes. W2 actual 14/estimate 12: both slow 1, error 2 minutes. Picture heights [4,7,9]/[5,6,10] cm: all errors 1 cm, MAE 1; categories 2/3 at >6, 3/3 at >7 with unchanged estimates.

  Category codes do not become quantities; zip assumes equal-length same-case order, units must match.

- Indoor is coded 1 and outdoor is coded 2. Someone averages **prediction**s and reports route type 1.5 as an accurate halfway category. Explain the problem and name an appropriate check.

> **My reasoning and evidence**
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> |---|
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> | |
> | |
> | |

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main duration pairs actual [8,10,14]/estimated [9,11,12], slow>10: classes [0,0,1]/[0,1,1], 2/3 match; errors [1,1,2] minutes, MAE 4/3≈1.33 minutes. W2 actual 14/estimate 12: both slow 1, error 2 minutes. Picture heights [4,7,9]/[5,6,10] cm: all errors 1 cm, MAE 1; categories 2/3 at >6, 3/3 at >7 with unchanged estimates.

  Category codes do not become quantities; zip assumes equal-length same-case order, units must match.

- A user needs the arrival duration in minutes, but a report gives only slow or not slow. What task should be specified, and what information was lost by the category **output**?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Pair the selected actual and estimated completion times in the stated order, using minutes throughout. Apply this task's own strict greater-than boundary, keep the per-case absolute errors and report the count before any fraction or mean. No model-fitting or broad performance evidence is supplied.

- For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes.

  Write actual and predicted category lists, count correct categories, calculate each **absolute error** and the mean error in minutes. State what each result measures.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Compare new evidence independently

> **My independent explanation, trace and limits**
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> |---|
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MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Confirm whether Plant **Class**ifier predicts a health category or a numerical severity score; document why.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Trace both checks for **Classification** and **regression**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Classification** and **regression**. Proposed explanation: “Numbers always mean **regression**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- System A predicts **class** codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Task A predicts healthy/spotted; B predicts next week's height in cm.
- Choose task and **output** types.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Task A predicts whether a plant is fern or grass. Task B predicts its height in centimetres. Which is **classification** and which is **regression**?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 3 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- System A predicts **class** codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes.

  Write actual and predicted category lists, count correct categories, calculate each **absolute error** and the mean error in minutes. State what each result measures.

My reasoning and evidence

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Distinguish predicting growth height from predicting a growth category.
- Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.
- All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.
- Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/3

Actual and estimated heights (paired order)



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/3

Classification or Regression?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Regression errors

Absolute errors are 1, 1, 1 cm; mean absolute error is 1 cm.



2. Actual categories

Actual classes are short, tall, tall.



3. Predicted categories

Estimated classes are short, short, tall: 2/3 correct.

CHANGE ONE THING

Change the category boundary to strictly greater than 7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant **Class**ifier
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Distinguish predicting growth height from predicting a growth category.
- Which labelled step matches this detail? Estimated **class**es are short, short, tall: 2/3 correct.
- 1. Actual categories
- 2. **Regression** errors
- 3. Predicted categories
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Predicted categories Estimated **class**es are short, short, tall: 2/3 correct. Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Change the category boundary to strictly greater than 7 cm. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Actual and estimated categories both become short, short, tall: 3/3 correct.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Actual and estimated categories both become short, short, tall: 3/3 correct. The numerical estimates did not improve; the decision definition changed.

Repair and keep the record

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

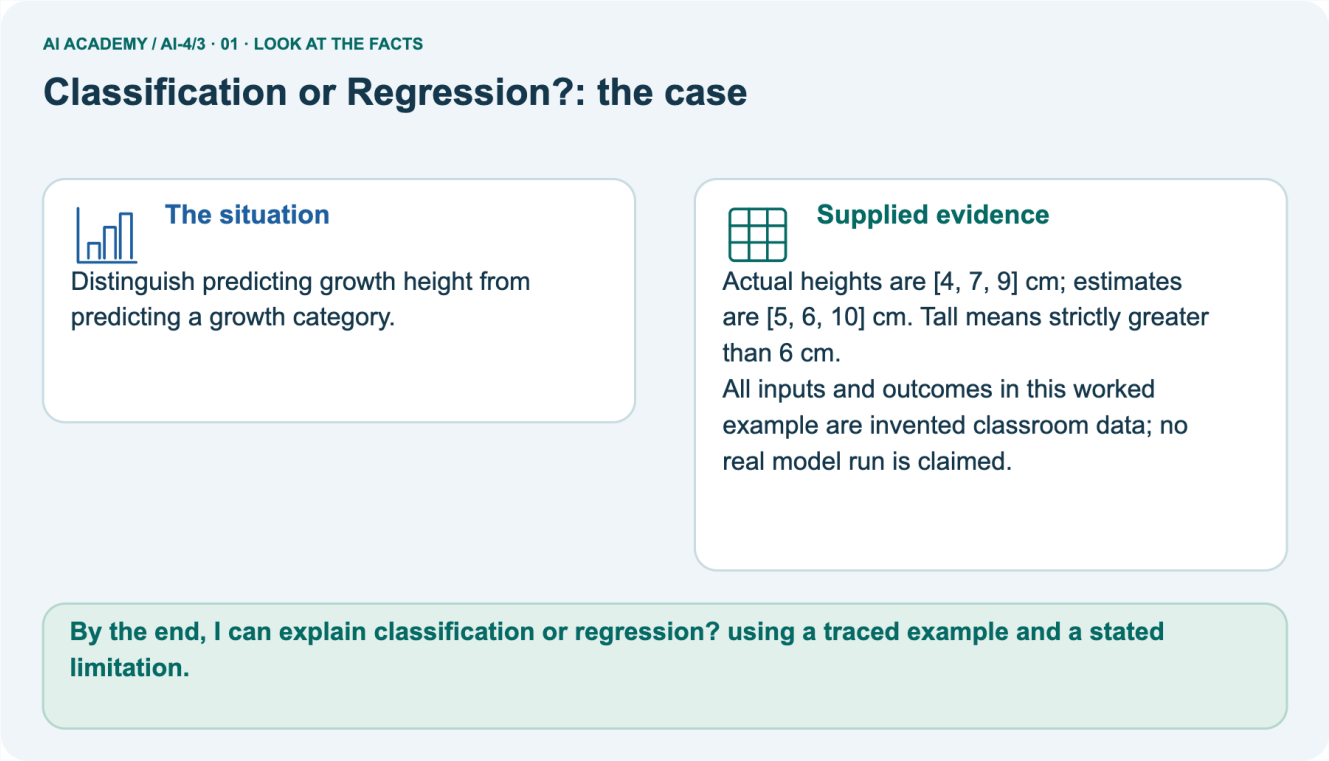
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Classification or Regression?: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

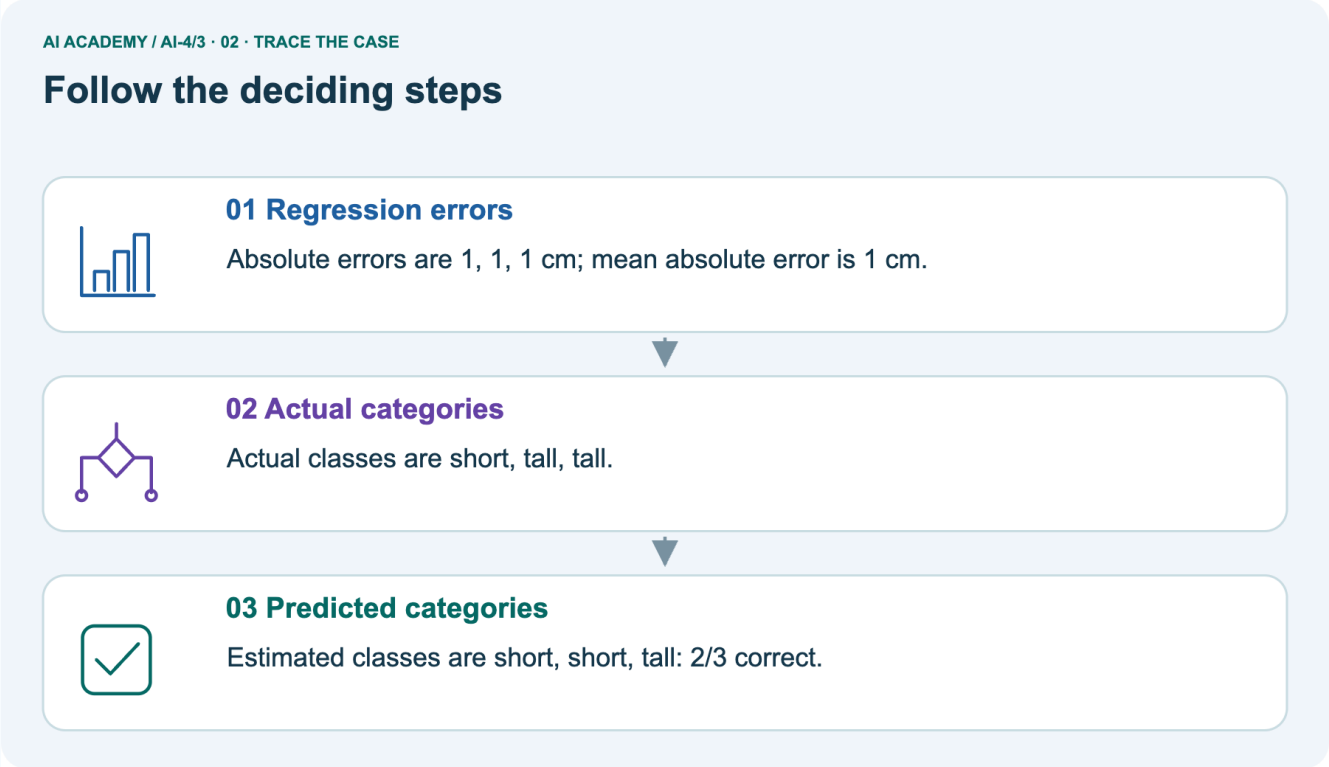
Distinguish predicting growth height from predicting a growth category. Supplied facts: Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Regression errors: **Absolute error**s are 1, 1, 1 cm; mean **absolute error** is 1 cm. → Actual categories: Actual **class**es are short, tall, tall. → Predicted categories: Estimated **class**es are short, short, tall: 2/3 correct.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Actual and estimated heights (paired order)



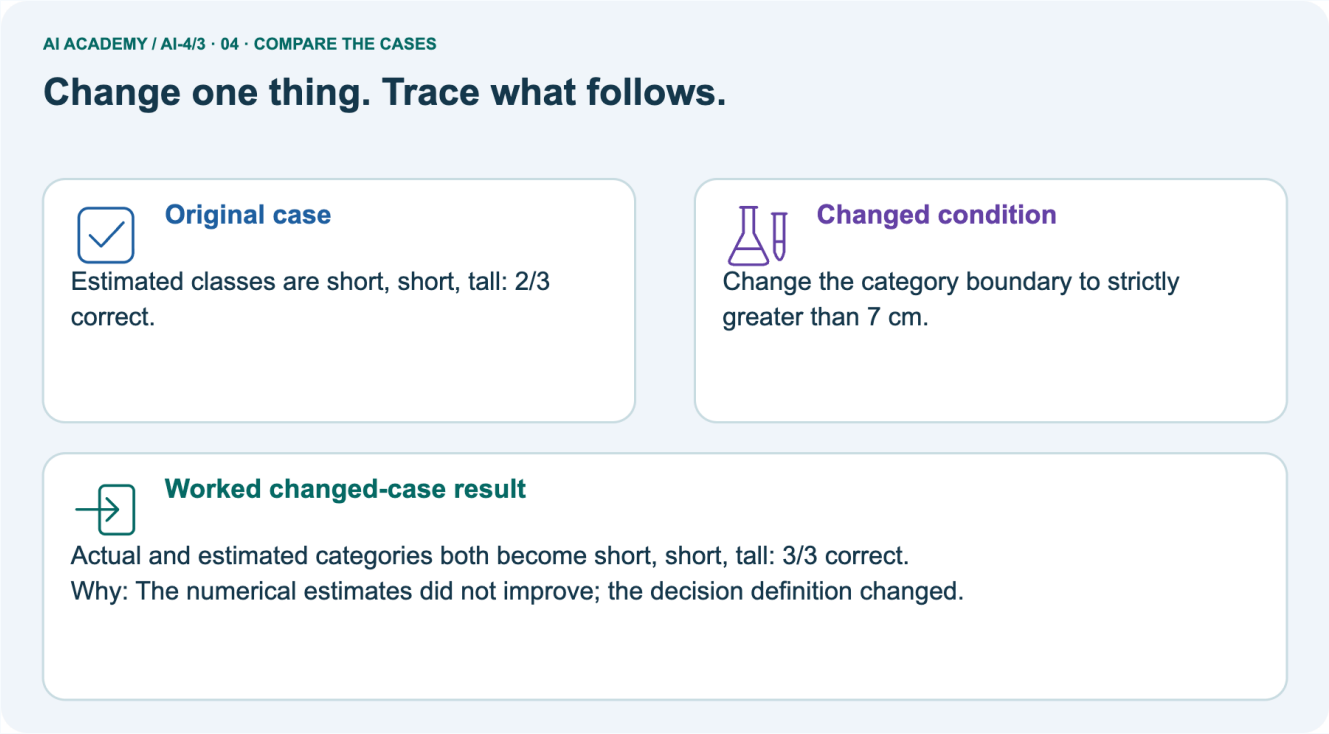
Follow the labelled picture with the matching explanation. Open the vector diagram

Actual and estimated heights (paired order). Units: cm. actual1: 4. estimate1: 5. actual2: 7. estimate2: 6. actual3: 9. estimate3: 10.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

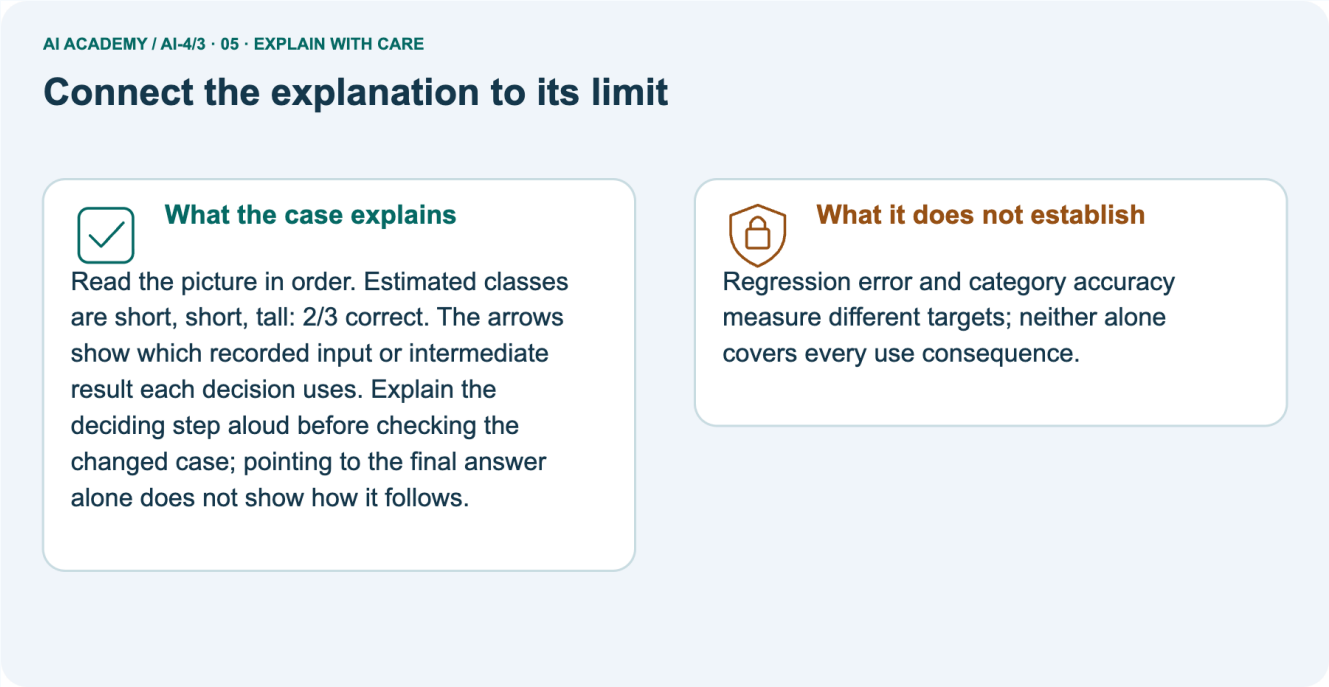
Original case: Estimated **class**es are short, short, tall: 2/3 correct. Changed condition: Change the category boundary to strictly greater than 7 cm.

Expected result: Actual and estimated categories both become short, short, tall: 3/3 correct. Reason: The numerical estimates did not improve; the decision definition changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct. The arrows show which recorded **input** or intermediate result each decision uses.

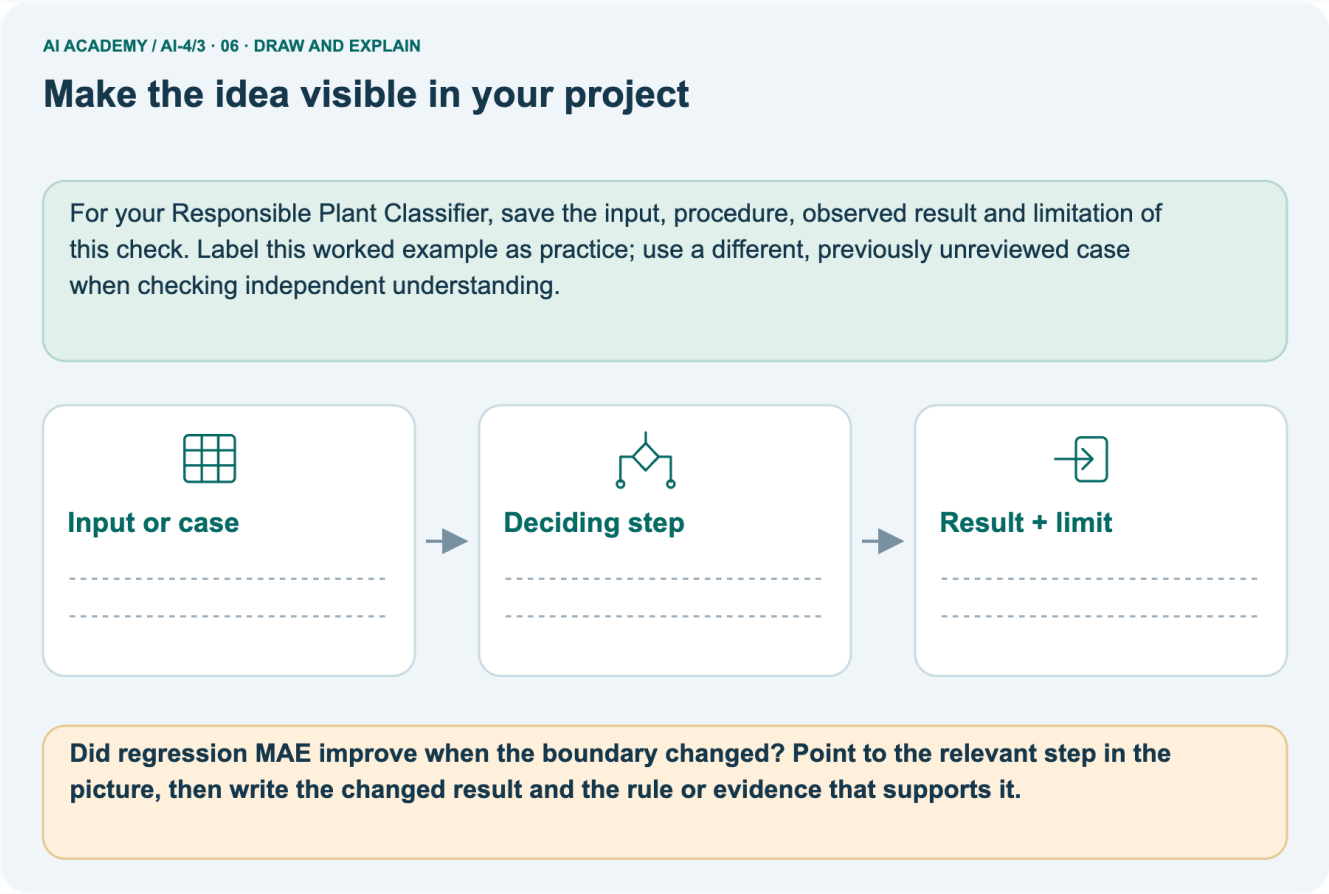
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Regression** error and category accuracy measure different targets; neither alone covers every use consequence.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Classification or Regression?: the case

02. Follow the deciding steps

03. Actual and estimated heights (paired order)

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.